

8 Magnetic Resonance Sounding

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8.1 Introduction

Magnetic Resonance Sounding (MRS) is a new technology which just passed the experimental stage to become a promising surface measurement tool to investigate directly the existence, amount and productiveness of ground water. The principle of Nuclear Magnetic Resonance (NMR), well known in physics, physical chemistry as well as in medicine, has successfully been adapted to assess the aquifer parameters.

The first high-precision observations of NMR signals from hydrogen nuclei were made in the forties of the last century. Meanwhile it is a standard investigation technology on rock cores and in boreholes (Kenyon 1992). The first ideas for making use of NMR in groundwater exploration from the ground surface were developed as early as the 1960s, but only in the 1980s was effective equipment designed and put to operation for surface geophysical exploration (Semenov et al. 1988, Legchenko et al. 1990). Extensive surveys and testing have been conducted in different geological conditions particularly in sandy aquifers but also in clayey formations as well as in fractured limestone and at special test sites (Schirov et al. 1991, Liebllich et al. 1994, Goldman et al. 1994, Legchenko et al. 1995, Beauce et al. 1996, Yaramanci et al. 1999, Meju et al. 2002, Plata and Rubio 2002, Supper et al. 2002, Vouillamoz et al. 2002, Yaramanci et al. 2002, Dippel and Golden 2003, Baltassat et al. 2005, Lange et al. 2005, Shushakov et al. 2005, Voillamoz et al. 2005).

8.2 NMR-Principles and MRS technique

The physical property used in NMR applications is the Spin of the nuclei under investigation, i.e. hydrogen protons of water molecules. This is to possess an angular momentum, without physically rotating, and an associated magnetic moment. The magnitude of this magnetic moment μ_0 , its equilibrium orientation and the oscillation frequency ω_L are related to the static field B_0 by:

$$\bar{\mu}_0 = \frac{\gamma^2 \hbar \bar{B}_0}{3kT} \frac{N}{V} \quad (8.1)$$

$$\omega_L = -\gamma B_0$$

where $\gamma = 0.267518$ Hz/nT is the gyromagnetic ratio for hydrogen protons, N/V is the number of Spins per volume and h , k , and T are common physical constants. By the application of a secondary electromagnetic field B_1 , perpendicular to B_0 and oscillating at the Larmor frequency ω_L , the spinning magnetic moment is deflected to an oscillating motion around the static field direction (Fig. 8.1a). During the excitation of the Spin it retains its magnitude and changes only the orientation describing a helix-shaped line on a unit sphere (Fig. 8.1b). The components of the magnetization vector perpendicular and parallel to the static field direction are harmonic functions of the magnitude of the secondary field B_1 and the pulse duration τ_p :

$$\mu_{\perp}(\tau_p) = \mu_0 \sin\left(\frac{1}{2}\tau_p B_1(\bar{r})\right) \quad \mu_{\parallel}(\tau_p) = \mu_0 \cos\left(\frac{1}{2}\tau_p B_1(\bar{r})\right) \quad (8.2)$$

Consequently the respective components of μ achieve arbitrary values between $+\mu$ and $-\mu$, and for long pulse durations or large magnitudes flip several times around. The perpendicular component continuously oscillates at the Larmor frequency in the plane perpendicular to B_0 , whereas the parallel component shows no such oscillating component.

According to the semi-classical solutions of the Bloch equations, describing the quantum mechanical phenomenon of NMR, both components of the magnetization decay individually and independently from their achieved magnitudes at pulse duration τ_p . The perpendicular component decays to zero with time constant T_2 , still precessing in the x-y plane, the parallel component builds up exponentially to its equilibrium magnetization with time constant T_1 . Consequently the magnitude of the total magnetization is not constant during the relaxation process. Its initial value at pulse cutoff is μ_0 and the final value after full relaxation is μ_0 again, but during the transient relaxation process its magnitudes varies with decreasing $\mu_{\perp}$ and increasing $\mu_{\parallel}$. The corresponding trajectory in a unit sphere for an excitation angle of 90° is shown in Fig. 8.1c. In general T_1 is $\geq T_2$, but at low static field strength, as given in Earth's field applications, they can be assumed to be equal.

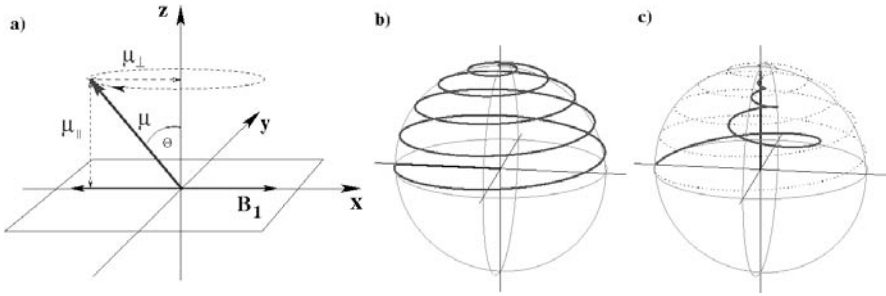


Fig. 8.1. a) Components of the precessing magnetic moment of the Spin. Unit trajectory of the Spin magnetic moment during the excitation process b) and during the relaxation process c) (after Levitt 2002). Direction of the static field B_0 is along the z-axis

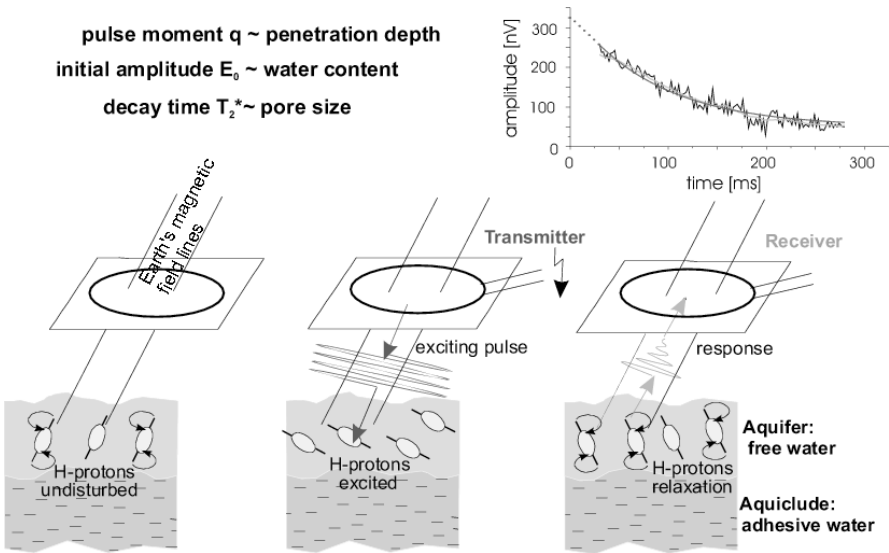


Fig. 8.2. Principle sketch of a Magnetic Resonance measurement with stages of protons undisturbed, excited and relaxing and relaxation signal

The relaxation follows the functions:

$$\mu_{\perp}(t) = \mu_{\perp}(\tau_p) e^{-\frac{t}{T_2}} \quad \mu_{\parallel}(\tau_p) = \mu_{\parallel}(\tau_p) \left(1 - e^{-\frac{t}{T_1}} \right) \quad (8.3)$$

The technique of Magnetic Resonance Sounding in geophysical applications makes use of the NMR phenomenon of the hydrogen protons of the groundwater (Fig. 8.2). The Spin system is perturbed by a pulse in the

transmitter coil and emits after pulse cutoff the response signal from the subsurface. Here, the Earth's magnetic field is the static field, leading to a Larmor frequency of some 1.2-2.5 kHz. The secondary field perturbing the Spin system is given by the electromagnetic fields of large surface loops. The response signal is also recorded in a surface loop, usually of the same dimension. In conventional depth soundings measurements are realized with transmitter and receiver being the same loop, the coincident configuration, but for 2D investigations both, transmitter and receiver are individual loops at arbitrary separation.

From a single recording the curve characteristics initial amplitude and decay constant of the transient curve are derived as well as the phase shift. By increasing the excitation intensity the depth sensitivity can be controlled.

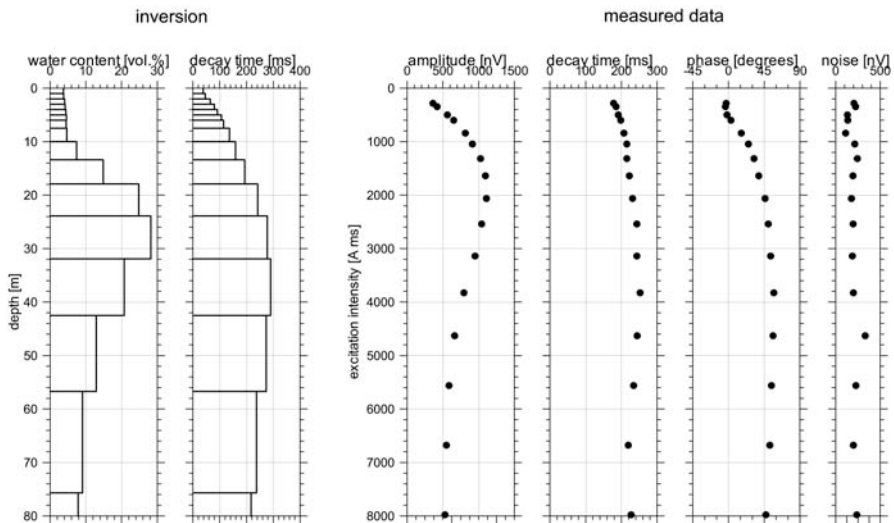


Fig. 8.3. Sketch of the pulse and signal registration sequence of a MRS recording (left). Data obtained after processing of the MRS recordings in dependency of the excitation intensity (right) and inverted subsurface parameters water content and decay time in depth (middle)

According to the previous explanation the tipping angle depends on the magnitude of the secondary field. Due to the nature of the transmitter loop field to decay rapidly in distance from the loop, Spins at shallow depths perform multiple rotations and by loss of coherence do interfere such that no resulting signal is emitted. In deeper layers the Spins achieve their maximum tipping angle and contribute the major part of the entire signal. A set of increasing pulses thus does not only provide an increasing penetra-

tion depth but also a depth selective focus. The thereby achieved sounding curve can now be interpreted by means of inversion to a water content and decay constant distribution with depth.

The basic equation describing the voltage response in the receiver loop in dependency of the excitation intensity, i.e. the pulse moment q , and the recoding time t , is given according to Weichman et al. (2000) by:

$$\begin{aligned} V_j(q, t) = & \omega \mu_0 \int_V d^3r f(r) \sin\left(-\gamma \frac{q}{I_0} |B_1(r)|\right) e^{-\frac{t}{T_2^*(r)}} \\ & \times \frac{2}{I_0} |B_1(r)| e^{i[\zeta_T(r) + \zeta_R(r)]} \\ & \times [\vec{b}_R(r) \cdot \vec{b}_T(r) + i \vec{b}_0 \cdot \vec{b}_R(r) \times \vec{b}_T(r)] \end{aligned} \quad (8.4)$$

Here, the pulse moment q is defined as the product of pulse duration τ_p and current through the loop I_0 , $q = \tau_p I_0$.

In this form the accounting terms of the equation can be physically interpreted as follows:

- The first line of the equation gives the signal amplitude of a Spin system, emitting the NMR response. It consists of the magnitude of the magnetization vector, i.e. the number of hydrogen nuclei, and the excitation angle they have achieved by the energizing pulse. The excitation angles of the resulting magnetization is a harmonic function, determined by the normalized amplitude of the co-rotating part of the transmitter field and the pulse moment q .
- The second line describes the sensitivity of the receiver to a signal in the subsurface. This spatial sensitivity is independent from signal generation, i.e. the pulse moment. It is simply given by the magnetic field distribution associated to the receiver loop. Additionally the signal undergoes a phase lag from the transmitter to the sample and from the sample to the receiver due to electromagnetic attenuation.
- The most important part in the treatment of the mathematical foundation of separated loop configuration is shown in the third line. Whereas the first two lines only cover the scalar values of excited signals and their associated induced voltage in the loop, this part accounts the vectorial nature of the evolution of the MRS signal. The expression in the bracket describes the dependency of the evolving signal on the directions of transmitter and receiver field and their orientation in respect to the Earth's field. Note, that only the unit direction vectors of the magnetic fields enter this part of the function. The first part of the sum, the real part of the whole bracket, is simply a scalar vector multiplication, whereas the second, the vector product of both fields and their scalar product with the Earth's field, scales the imaginary part of the expression. In

case of coincident loop soundings, the latter part vanishes whereas the former one approaches unity; the complete expression becomes a real unity value. For all other loop layouts the value of this last term, the geometric term, scales the NMR effect and the signal response by a value in the range $[-1, 1 ; -i, i]$. Consequently, the evolving signal for separated loops is in general complex valued. The resulting phase shift of the recorded signals is the geometrical phase of the MRS signal. It acts in addition to the quantum mechanical phase due to frequency deviations (Hertrich and Yaramanci, 2003) and frequency spectra of the pulse (Legchenko, 2004) and the electromagnetic phase lag caused by signal propagation in conductive media.

The water content entering this equation is only part of the total water content in the pore space. Water that is bound to the internal surface or stored in very fine pores does not contribute to the MRS response signal since its relaxation constants are too small to be detected by the current MRS technique. The water content contributing to the response signal is thus closely related to the mobile water content. The remaining parameters in the integral of Eq. 8.4 then describe the constant settings of the measurement as loop shape, Earth's electrical conductivity and choice of pulse moments and are commonly combined to the MRS kernel function:

$$V_R(q, t) = \omega_L \mu_0 \int_V d^3r f(r) K(q, t) e^{-\frac{t}{T_2^*(r)}} \quad (8.5)$$

$$K(q, t) = \sin\left(-\gamma \frac{q}{I_0} |B_1(r)|\right) \times \frac{2}{I_0} |B_1(r)| e^{i[\zeta_T(r) + \zeta_R(r)]} \times [\vec{b}_R(r) \cdot \vec{b}_T(r) + i \vec{b}_0 \cdot \vec{b}_R(r) \times \vec{b}_T(r)]$$

In case of 2D or 1D conditions, the formulation is commonly simplified by reducing the kernel function to the necessary dimension by integrating the general kernel in direction of the respective Cartesian dimension. The expression for 3D is then

$$V_R(q, t) = \int_0^\infty \int_{-\infty}^\infty \int_{-\infty}^\infty f(x, y, z) K_{3D}(x, y, z) e^{-\frac{t}{T_2^*(x, y, z)}} dx dy dz \quad (8.6)$$

and can be reduced to 2D as a section in depth and profile direction like

$$\begin{aligned}
 V_R(q, t) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, z) K_{2D}(q; x, z) e^{-\frac{t}{T_2^*(x, z)}} dx dz \\
 K_{2D}(q; x, z) &= \int_{-\infty}^{\infty} K_{3D}(q; x, y, z) dy; \quad \partial f(y) / \partial y = 0
 \end{aligned} \tag{8.7}$$

Reducing the kernel to 1D allows only a water content variation in depth

$$\begin{aligned}
 V_R(q, t) &= \int_0^{\infty} f(z) K_{1D}(q; z) e^{-\frac{t}{T_2^*(z)}} dz \\
 K_{1D}(q; z) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K_{3D}(q; x, y, z) dx dy; \quad \partial f(x) / \partial x = \partial f(y) / \partial y = 0
 \end{aligned} \tag{8.8}$$

The principle how the multiplication of a water bearing layer and the kernel function leads to a characteristic sounding curve is shown in Fig. 8.4. Here the 1D kernel function according Eq. 8.8 for selected pulse moments is represented by line plots. The shaded area below the curves of the kernel function represents the resulting signal amplitude.

The characteristics of the kernel function by means of depth sensitivity is significantly determined by the loop size. Fig. 8.5 shows the kernels for loop sizes of 100 m, 50 m and 20 m above an insulating halfspace as contour plots. The characteristic depth focus changes to shallower depth for smaller loops. However, it is visible that the penetration depth can not be arbitrarily increased by only increasing the pulse moment since the focus depth converges to a maximum depth. Due to the reduced investigated volume for smaller loops the magnitudes of the kernels and consequently the maximum possible response signals decrease with loop size and currently limit the method for very shallow applications in the range of the usually observed natural noise.

A high potential of the technique of separated loop MRS measurements lies in the investigation of 2D structures. Similar to the previous formulation of the 1D kernel, the general formula for the MRS response is reduced by summation according Eq. 8.7 along one dimension. The resulting 2D kernel then represents the sensitivity of a sounding to a 2D water content distribution, assumed to be infinitely extended in the summation direction. In contrary to the depth focus of the coincident soundings in Fig. 8.6, separated loops soundings provide a sensitivity to shallow structures in the range of the loops. This pattern is more prominent for increasing loop separation. The combination of several soundings along a profile with different loop separations consequently provides a high coverage of 2D sensitivity and allows the derivation of a 2D water content distribution from a profile like survey.

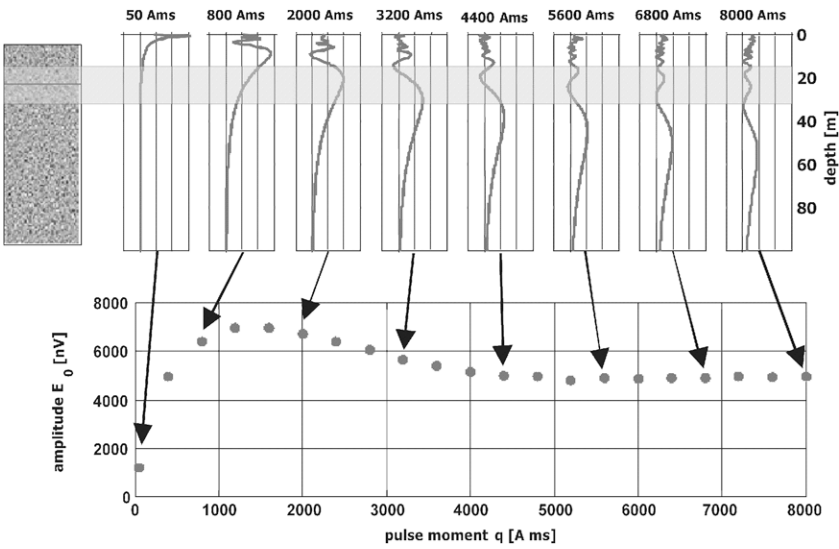


Fig. 8.4. Kernel functions (i.e. sensitivities) for different excitation intensities (pulse moments) and constitution of a sounding curve for amplitudes

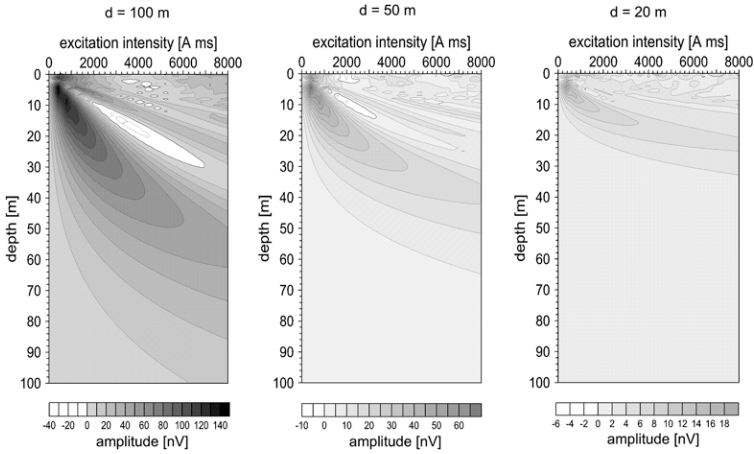


Fig. 8.5. Kernel functions (for 1D soundings) for different loop sizes

Such a reconstruction of the physical parameters from a set of measurements with overlapping and variable coverage of a subsurface section makes it, according the generally accepted convention in geophysics, a tomography application, e.g. geoelectrics, and therefore we call it Magnetic Resonance Tomography (MRT).

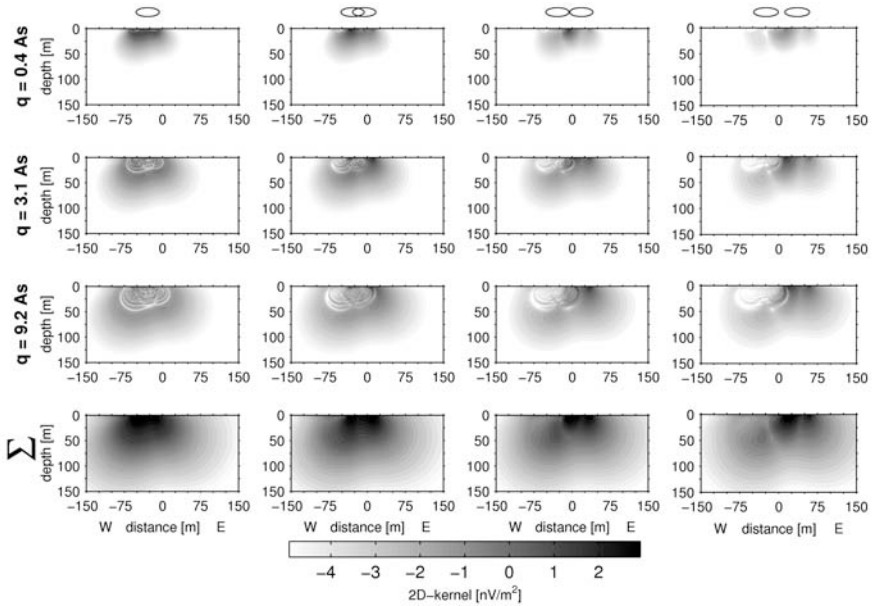


Fig. 8.6. 2D kernels for increasing loop separation for three selected pulse moments (lines 1-3) and as the cumulative kernels for an entire sounding. Measurement configuration is 48m loop diameter, 2 turns, Earth's field at 48000 nT and 60°inclination, profile direction 45°N. Kernel magnitude is in logarithmic scale; dark represent higher sensitivities

8.3 Survey at Waalwijk / The Netherlands

The geology of the area mainly consists of well-sorted sands. At this site, a borehole nearby confirms a shallow aquifer with a water table at about 6 m. The lower boundary of the aquifer is at 52 m depth to clay and silt units (Lubczynski and Roy 2003). All units are heterogeneous and consist of thin layers.

The electrical subsurface model at the site, as derived from TDEM measurements shows moderate to low ground resistivities, thus reducing the effective penetration depth of the measurements. Hereby, the low resistivity values correspond to clayey material between 52 and 80 m depth. The local geomagnetic field intensity during the survey was 48360 nT (corresponding to a Larmor frequency $f_L=2059 \text{ Hz}$) with an inclination of approximately 60 degrees.

Table 8.1. Electrical subsurface model as derived from TDEM and lithology as derived from a nearby borehole for the site Waalwijk.

depth [m]	resistivity [Ωm]	lithology
0 – 6	120	fine sands
6 – 52	60	fine – coarse sands
52 – 85	20	clayey sands
> 85	60	medium sands

The measurements were conducted using a square loop with a side length of $a = 113.5$ m. Excitation intensities (24 pulse moments) ranged from 57 to 5350 A ms. Considering the resulting MRS sensitivity for this configuration a reliable depth of investigation of approximately 60 m can be expected. The total recording length of the decaying signals is 450 ms with a 'dead time' between the exciting pulse and start of recording of 40 ms. The data generally show a high S/N ratio (example for pulse moment $q = 500$ A ms plotted in Fig. 8.7). Therefore, the relaxation data is of good quality even for lower amplitudes, i.e. later recording times.

MRS relaxation signals can exhibit a multi-exponential relaxation behaviour (Fig. 8.7) due to a signal superposition of layers having different decay times ($\sim$ grain sizes) and / or due to a possibly multi-modal decay time distribution within the layers. Whereas the signal amplitudes of MRS are proportional to the amount of water in the subsurface, the decay times are a function of grain sizes (e.g. Shirov et al. 1991, Yaramanci et al. 1999) and pore sizes (e.g. Kenyon 1992). While small grain sizes, i.e. pores, correspond to small decay times large grain sizes, i.e. pores, will exhibit long decay times.

The so far available inversion schemes make use of an approximated mono-exponential fitting approach of the NMR relaxation. However, this practice does not take into account that, analogous to borehole and laboratory NMR, every signal contribution of a single sample or volume element is a superposition of signal contributions due to a specific grain size i.e. pore size distribution within the investigated sample / volume element in the subsurface. Therefore, a newly developed inversion scheme with regard to the multi-exponential nature of decay in the layers and that of the recorded signal too (Mohnke and Yaramanci 2005) has been used by the inversion.

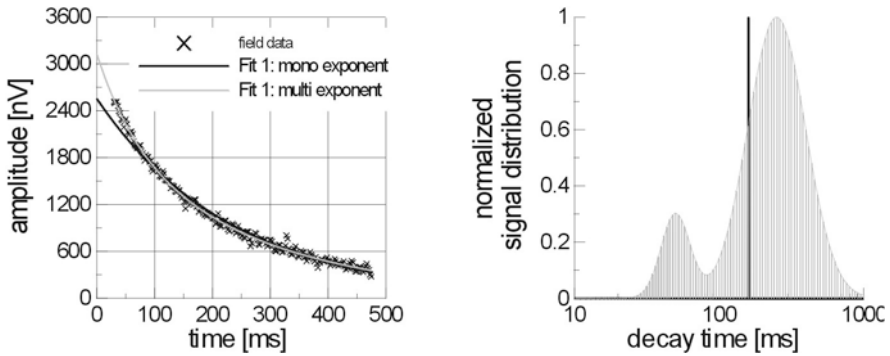


Fig. 8.7. Example of measured MRS relaxation data (crosses) at Waalwijk-2 with mono-exponential (black line) and multi-exponential (grey line) fitting of the data and corresponding decay time distributions (right)

Inversions have been carried out using both conventional mono-exponential inversion (Fig. 8.8a) and multi-exponential inversion approaches using free (Fig. 8.8b) and fixed decay time distributions (Fig. 8.8c). Mono-exponential fitting shows a shallow aquifer between 6 and 50 m with a mean integrated water content of about 28 %. The inverted (mono-exponential) decay times range from 200-300 ms for the aquifer layers and is about 100-150 ms for neighbouring layers. For greater depths (> 50 m) the decay times increase to values of about 400 ms coinciding with an increase of data misfit and do not comply with the fine grained material expected from borehole data.

Multi-exponential inversion results (Fig. 8.8 b,c) show an improved data fit, especially at the beginning of the signal (crosses) as the consequence of a better fit of the short decay times. The integrated water content, being about 34 % within the aquifer, is generally higher. The multi-exponential decay time distribution within the aquifer layers show that the main fraction of the water content can be associated to prominent decay times around 400 ms, (medium to coarse material). In addition a small fraction of the water content is associated to rather small decay times between 40 – 60 ms and consequently resulting in higher integrated water content when extrapolating to initial amplitude ($t = 0$ ms) values (see also Fig. 8.8 left). This may indicate small intrusions of clayey / fine grained material or more likely be of an artificial origin occurring due to low S/N for small q values that focus on these more shallow depths. For depths below 50 m the decay time distribution indicate a different type of lithology. Here, decay times of about 90-100 ms (while slightly overstated for pure clay and silt with 50 ms) reflect the clayey / fine-grained material as expected from borehole measurements.

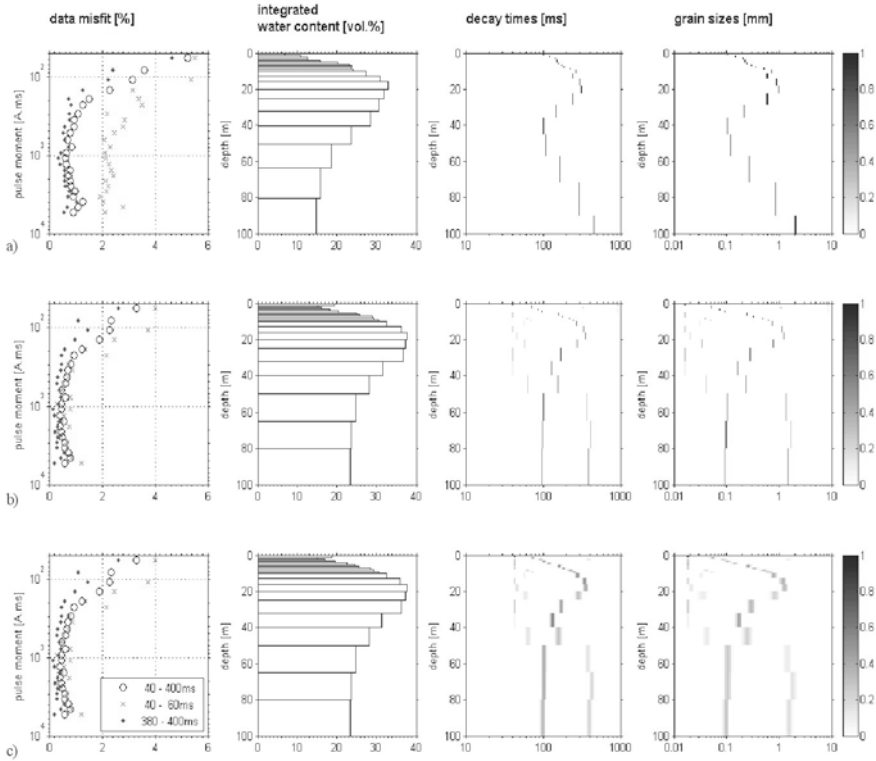


Fig. 8.8. Inversion results for the site Waalwijk-2. a) with conventional mono-exponential inversion, b) with multi-exponential inversion using two free decay times, c) with fixed decay times. Left: Percentage data misfit for recorded relaxation signals (windows: total [o] start [x] and end [.]); Center left: Integrated mobile water content; Center right: Decay time distributions; Right: Derived grain size distributions. The grey scale gives the fractional water volume in the material with corresponding decay times and grain sizes

However, besides these prominent maxima at low decay times a distinct, albeit smaller, maximum appears at decay times of about 450 ms. This may be an indication of interbedded layers of coarse material within the layers, that cannot be discriminated in detail due to a decreasing resolution of the inversion for these depth. This would correlate to the heterogeneous layering derived from the borehole.

Another possibility for this bi-modal distribution may be just a ‘shadow’-effect from the high decay times of the aquifer. Fig 8.8 at right shows grain size distributions as derived from MRS decay times according to an empirical relation introduced by Shirov et al. (1991) and Yaramanci et al. (1999). While the aquifer shows grain sizes in the range of 1 mm

(coarse sands, gravel) the grain sizes < 0.1 mm below 50 m represent the fine material of the aquiclude.

While the conventional approximated inversion approaches a priori assume mono-exponential MRS relaxation data, a comprehensive inversion scheme allows to take into account the multi-exponential features.

Whereas conventional inversion will reduce the resolution of MRS using more or less averaged decay times, comprehensive inversion can yield a more realistic decay time distribution with depth. Moreover, since the water content is proportional to the initial signal amplitude ($t=0$ ms) the significant improvement of data fitting by multi-exponential inversion of MRS signals is basic for an improved quantitative determination of the integrated mobile water content in the subsurface, that so far is usually underestimated by the conventional approach.

8.4 Survey at Nauen / Germany with 2D assessment

The survey in Nauen was performed on a well investigated test site, where a variety of additional data are available (Yaramanci et al. 2002). From previous surveys a shallow aquifer from about 2 m down to some 20 m is known, that slowly crops out towards the surface. From geoelectrical and radar sections the upper boundary of the clayey confining bed is known. However, the deeper structures are neither visible in radar-section nor can they be interpreted by a resistivity contrast. The MRS survey was designed to cover the outcrop of the aquifer and spread towards the as aquiclude assumed direction. There, previous individual MRS measurements did already point to larger water contents than they were presumed from geoelectrics and geological conditions.

The survey was performed (Hertrich 2005) with 4 individual positions of loops with 48 m diameter, adjacent half overlapping. All sixteen possible permutations of transmitter and receiver positions have been realized. The result of various inversions is shown in Fig. 8.9 and the corresponding measured data set is shown in Fig. 8.10.

In the sounding data matrix on the right hand side the modelled data are plotted (Fig. 8.10) in comparison to the measured data. The solid lines show the data adaptation with all 16 soundings. The dashed lines are calculated from the model that has been reconstructed by using only the 4 coincident soundings. The deviation between both therefore shows the gain in model resolution by fitting all available data. The found model (Fig. 8.9 3. from bottom) represents the presumed outcropping aquifer on the lefthand side around P1. Its boundary obviously crops out at around P2 with an

immediate transition into a second undulating water bearing layer towards right side. Both structures have their lower boundary at a layer of very low water content at some 20 m. Below this zone a second aquifer appears at some 40-50 m depth.

The lower boundary of the low water content zone shows a significant depression at around +20 m on the profile. The trisection of the section into a first aquifer, confining bed and second aquifer can be roughly estimated from the 2D inversion using only the coincident soundings as shown in the second model in the left hand side of the figure. A resolution of a depth variation of layer boundaries or a discrimination of structures within the layers is, however, not given. The single 1D inversions and the pseudo-2D plot (Fig. 8.9 bottom and above) do also just roughly show the structure of a low water content layer between two aquifers but does not reflect the structure of this body as it is rendered by full 2D inversion of the complete dataset.

Comparing the uppermost 30 m of the complete inversion result of the MRS survey including all available data to the inversion result of a resistivity section in Wenner configuration of 2 m basic electrode spacing allows a combined interpretation. The top layer, showing water contents close to zero is represented by resistivities of some 4000 Ωm from the left to P3 and less than 100 Ωm from P3 to the right. This corresponds to a dry sand layer at the left changing to clay material of the outcropping till layer to the right. The first aquifer, intersected at around P2 has water contents of about 35% in the left area and slightly less in the right part. However, only the left part is represented by increased resistivities from the resistivity of the surrounding clayey material. From the vanishing difference in resistivities of the right part of the aquifer a significant clay content and a corresponding reduced hydraulic permeability can be assumed here.

From the conventional resistivity investigation, the aquifer system at this location would not have been characterized correctly due to the ambiguous correlation of resistivities and lithology. Here, the definition of aquifer structures from MRS and the combined interpretation of the subsurface properties by resistivity and water content leads to a reliable estimation of aquifer characteristics and delineation of clayey materials.

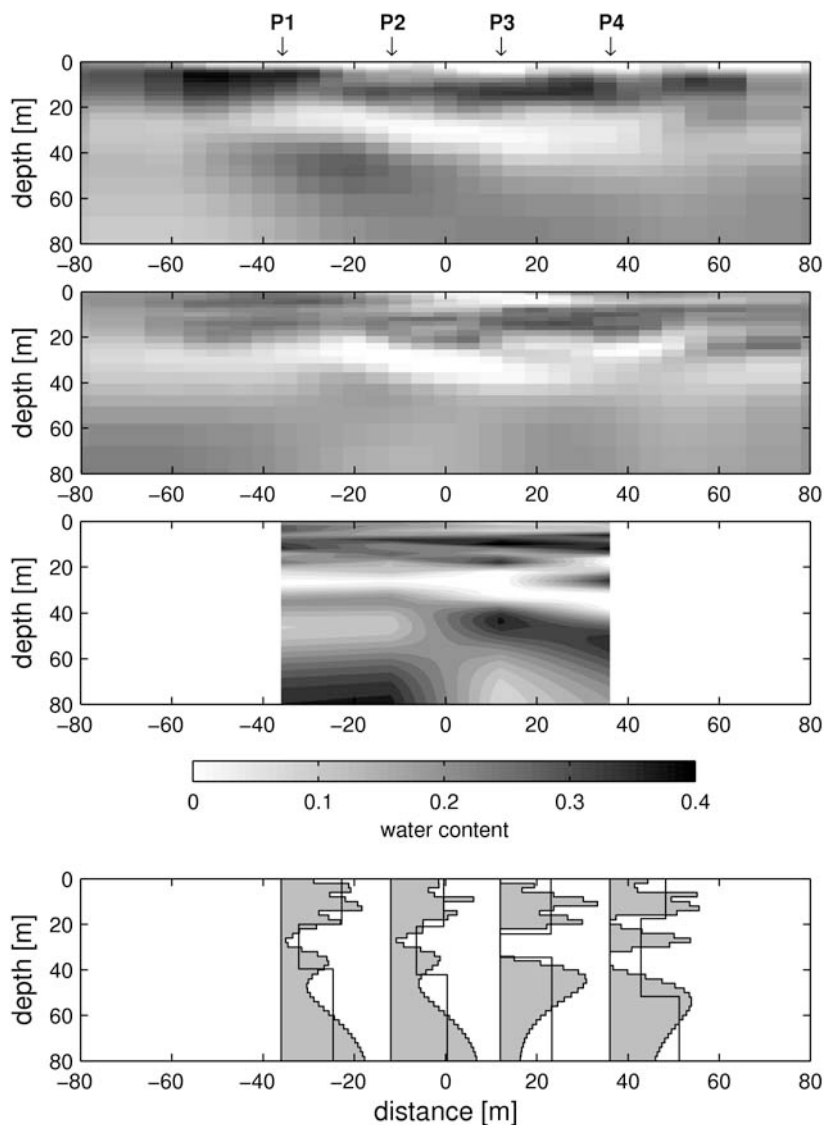


Fig. 8.9. Results of the 2D survey in Nauen by different inversion approaches. From bottom to top: 1) Inversion of coincident loop measurements (smooth and 3 layer block inversion), 2) contours of smooth inversion in 1), 3) Joint inversion of coincident loop measurements 4) Joint inversion of coincident loop and separated loop measurements

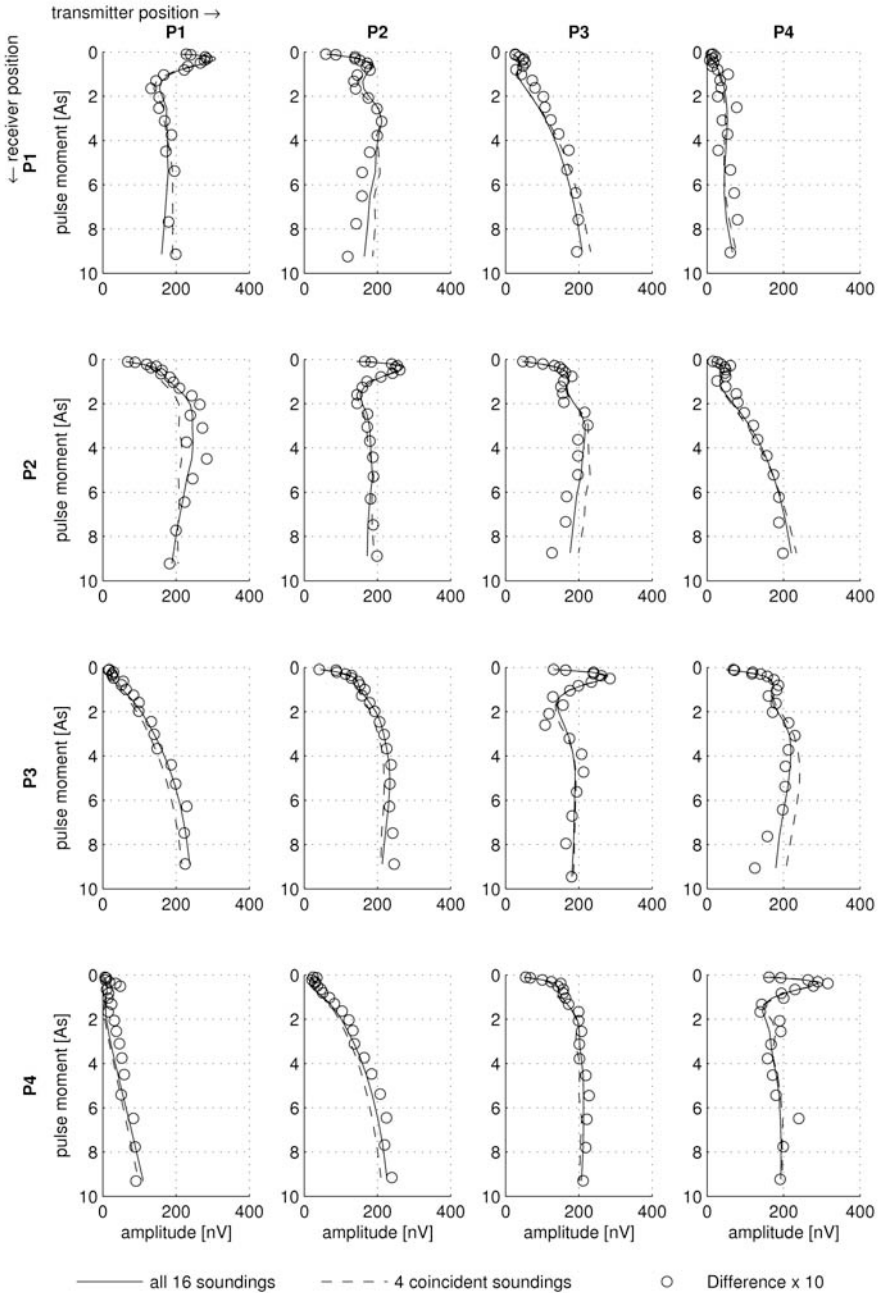


Fig. 8.10. Measured amplitude data and modelled curves for the inversion results (in Fig. 8.9) for joint inversion of coincident loops and for joint inversion of coincident and separated loops (full data)

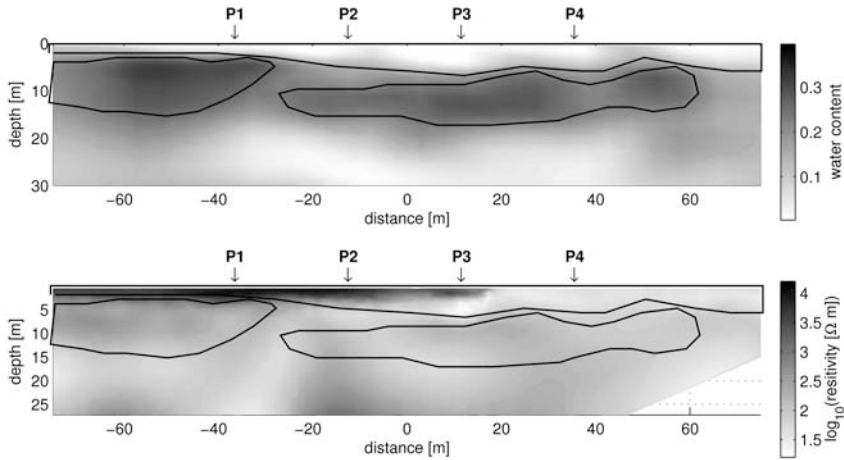


Fig. 8.11. Comparison of MRS and geoelectrics at the test site Nauen. Distinguishable structures in water content in combination with their electrical pattern allow a more profound interpretation of hydraulic properties

8.5 Current developments in MRS

The method of SNMR has just passed the experimental stage to become a powerful tool for groundwater exploration and aquifer characterization. Some further improvements are still necessary and just in work.

The most concern currently is the effect of resistivities and their appropriate inclusion into the analysis and inversion. In MRS the exciting field will be distorted and polarized considerably due to induction in the presence of conductive structures. Earlier considerations of this (Shushakov and Legchenko 1992, Shushakov 1996) have led to new improved theoretical description and numerical handling of this problem (Weichmann et al. 2002, Valla and Legchenko 2002). For an appropriate analysis of MRS the resistivity structure should always be available by which geoelectrical method ever and incorporated into the analysis. The experience shows that even moderate conductive structures may have large distortion effects.

In fact the incorporation of resistivities allows an improved modelling of the complex MRS signal, i.e. the phases, (Braun et al. 2002, 2005) which is not only useful for understanding the measured phases but is also the basis for a successful inversion of complex signals to yield the resistivity information directly from SNMR measurements.

In the analysis of SNMR generally the relaxation is assumed to be mono-exponential. Even if individual layers show a mono-exponential decay, the integration results in a multi-exponential decay in the measured signal. The most comprehensive way taking account for this is to consider a decay time spectra in the data as well as in the inversion as shown in Chap. 8.3. This leads to the pore size distribution which is a new information and allows improved estimation of hydraulic conductivities. The determination of water contents will also be significantly improved as the initial amplitudes are much better fitted using a decay spectra approach.

Currently SNMR is carried out mainly with a 1D working scheme using coinciding loops. The 2D surveying as shown in Chap. 8.4 is not standard yet. The errors might be very large by neglecting the 2D or even 3D geometry of the structures (Warsa et al. 2002, Hertrich et al. 2005) which have to be considered in the analysis and inversion in the future. Measurement layouts are to be modified to meet the multi-D conditions, which is easier to accomplish for nonconductive structures. In multi-D structures the actual difficulty is the numerical incorporation of the electromagnetic modelling for the exciting field even when the electrical data is available. The full 2D inversion for decay times are not available yet.

As in any geophysical measurement also with SNMR the inversion plays a key role by interpreting the data. The limits of inversion and also the imposed conditions in terms of geometrical boundary conditions as well as differences in the basic physical model used may lead to considerable differences. The inversion of SNMR data may be ambiguous, since not only different regularizations in the inversion impose a certain degree of smoothness upon the distribution of water content (Legchenko and Shushakov 1998, Yaramanci et al. 1998, Mohnke and Yaramanci 1999) but also the number of layers and the size of layers forced in the inversion may considerably influence the results. The rms-error is not necessarily a sufficient measure for assessing the quality of the fit of a model to the observed data. The most recent research suggests that a layer modelling with free boundaries avoids the problems associated with regularization and takes into account the blocky character of the structure where appropriate (Mohnke and Yaramanci 2002). Similar consideration are needed for 2D inversions considering the imposed cell sizes and shapes as well as smoothing constraints.

Further improvement in the inversion can be achieved in case of geoelectrical measurements are available and they can be incorporated into a joint inversion with SNMR. Examples of joint inversion of SNMR with Vertical Electrical Sounding show considerable improvement in the detectability and geometry of the aquifers and allows also by utilizing of ap-

appropriate petrophysical models the separation of mobile and adhesive water (Hertrich and Yaramanci 2002).

Finally the importance of the SNMR method lies in its ability to detect water directly and allowing reliable estimation of mobile water content and hydraulic conductivity. In this respect it is unique, since all other geophysical methods yield estimates, if ever, indirectly via resistivity, induced polarization, dielectric permittivity or seismic velocity. Using SNMR in combination with other geophysical methods not only allows direct assessment but is complementary to the information yielded by other geophysical methods.

8.6 References

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